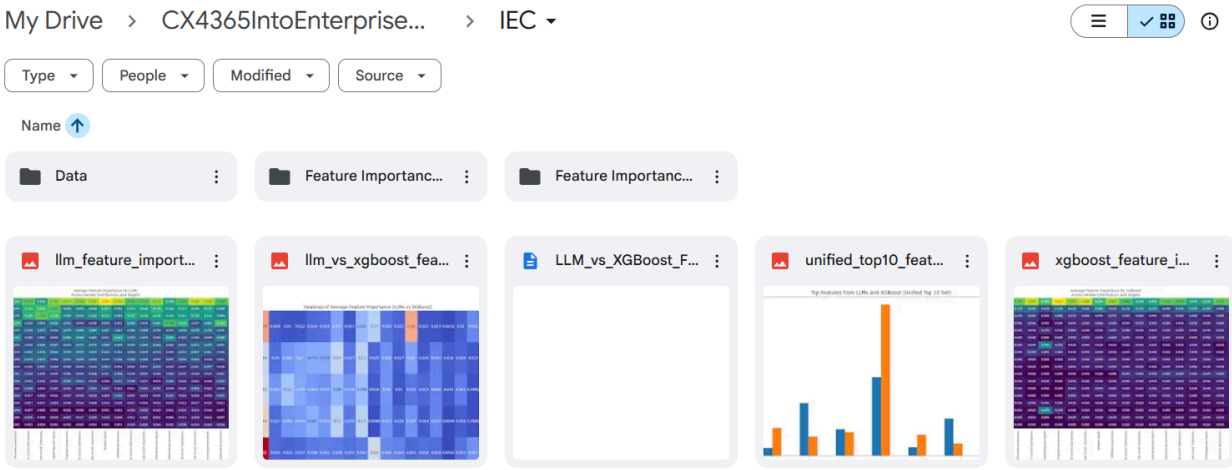
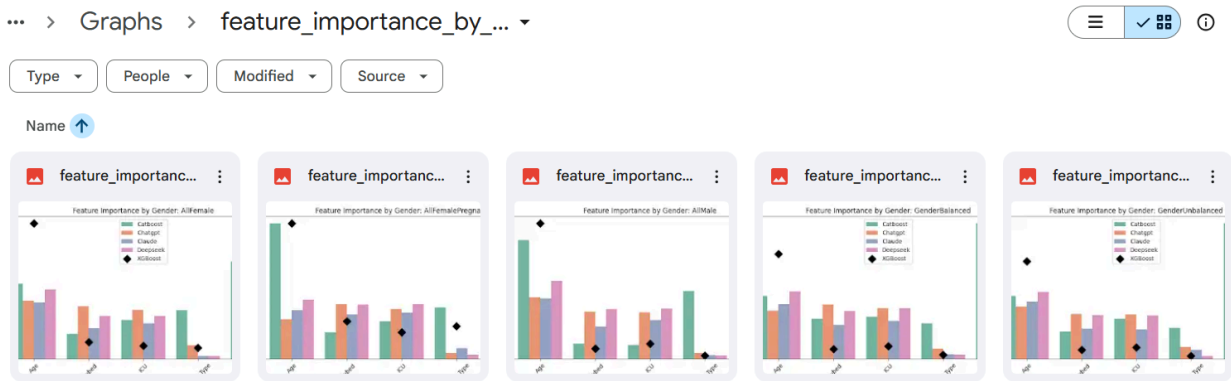
Result Reproduction:





LIKE Reproducibility Exercise

Project Plan

The main goal of this project was to compare how different types of AI models try to predict COVID 19 severity based on which different medical features they deemed as important. Specifically, we wanted to see whether the comparison of traditional Machine Learning models and new “Black Box” LLMs look at the same things when producing their outcomes. The design involves a large dataset of over 1 million patients that are run through each model to get scores based on which features were more or less important. These were then compared amongst each other using bar charts and heatmaps. To also ensure biases were not included, the models were also given the data by gender groups to determine if their rankings would change.

Project Execution

The team first cleaned the large datasets and translated each row into understandable sentences so the LLMs could read the information properly. They also ensured to use multiple models to have a more controlled environment. This consisted of three major LLMs and two different traditional models. They also performed “prompt engineering” to tweak the AI questions so the LLMs were outputting usable numbers instead of random fluff text.

Their results showed a huge difference in how these AIs think. It was agreed upon by the LLMs and traditional models that Age was the highest risk factor. However, every other medical feature was disagreed upon. The LLMs focused on more technical and clinical information as if they were learning directly from medical textbooks. This made information such as if a patient was intubated important for LLMS. On the other hand, the traditional models focused more on administrative data, that made information such as the medical , important. It was also found that LLMs were more consistent across genders, whereas traditional models were more unstable when looking at specific groups.

Supporting Evidence of the claimed Project Execution

After running the simplified code using Google Colab, I was able to verify these claims from their report using the outputs. Using the bar graph “Top Features from LLMs and XGBoost (Unified Top 10 Set),” we are able to see similarities and differences based on how significantly far apart the orange and blue bars are. This shows that there is a larger gap in favor of LLMs regarding their importance for Intubed patients and a larger gap in favor of XGB regarding their importance in placed Medical Units. Using the heatmaps, we are able to prove the claim that Age was the top predictor for all LLMs and XGBoost. This is shown using the heatmap as the highest valued and warmest color of each model is within the Age column. Lastly we can help support the claim that classical models varied more by gender. Using the Faceted Gender Charts, it showed the XGBoost diamonds and CatBoost bars to jump around between charts. On

the other hand, the three LLMs typically stayed flat which aligns that LLMs were mor consistent across genders.

Project Evaluation

- **Plan (120%):** Their plan was quite large. They not only grabbed a dataset and ran a default model, but they also built whole systems to convert data into prompts, test different gender scenarios, and repeated the process for five different AI models. It goes way beyond what a project would require in a standard class.
- **Match (120%):** They accurately built everything they had mentioned they would in the proposal. They also handled the cleaning, prompt engineering, and statistical comparisons well while creating different ways to present the data. They even dealt with very specific scenarios, like including Pregnant Females, instead of ignoring those outliers.
- **Factual (100%):** The results in their report are factual and easily reproducible. When I ran the code myself, I was able to extract the same splits they described in their report. The graphs I generated also looked just like the patterns they described in their analysis. Hence, their claims were substantial.